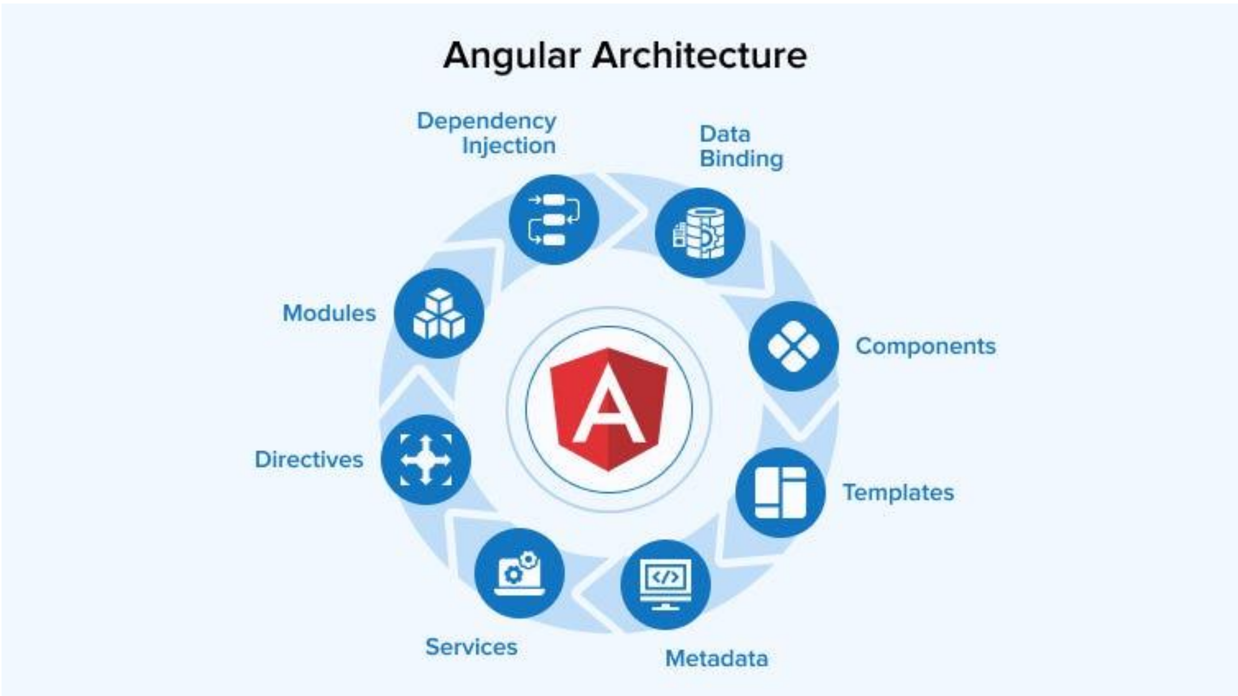
1. Angular Overview & Its Versions



What is Angular?

Angular is a **TypeScript-based front-end framework** developed and maintained by **Google** for building **Single Page Applications (SPAs)**.

Key characteristics:

- Component-based architecture
- Strongly typed using TypeScript
- Built-in support for routing, forms, HTTP, and dependency injection
- Enterprise-ready and scalable

Why Angular?

- Two-way data binding
- Modular architecture

- High performance with Ahead-of-Time (AOT) compilation
- Excellent tooling (Angular CLI)
- Widely used in enterprise applications

Angular vs AngularJS

AngularJS	**Angular**
JavaScript	TypeScript
MVC	Component-based
Poor performance	Optimized & fast
Deprecated	Actively supported

Angular Versions (High-Level Timeline)

Version	**Highlights**
Angular 2	Complete rewrite
Angular 4–7	Performance & tooling improvements
Angular 8	Ivy preview
Angular 9	Ivy default renderer
Angular 10–12	Stability & DX improvements
Angular 13	No View Engine
Angular 14	Standalone components
Angular 15	Improved standalone APIs
Angular 16	Signals introduced
Angular 17	New control flow (@if, @for)
Angular 18 (latest)	Performance, SSR, DX improvements

Angular follows **semantic versioning** and **LTS support** for enterprise stability.

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2. Prerequisites for Angular

Before learning Angular, you should be comfortable with:

Mandatory Skills

- **HTML**: elements, attributes, forms
- **CSS**: flexbox, basic layouts
- **JavaScript (ES6+)**:
  - let, const
  - Arrow functions
  - Promises
  - Array methods (map, filter)
- **TypeScript Basics**:
  - Types
  - Interfaces
  - Classes
  - Decorators

Tooling Knowledge

- Node.js
- npm (Node Package Manager)
- Basic CLI usage (command prompt / terminal)

Since you are already working with **Angular CRUD + API integration**, these prerequisites align directly with real-world usage.

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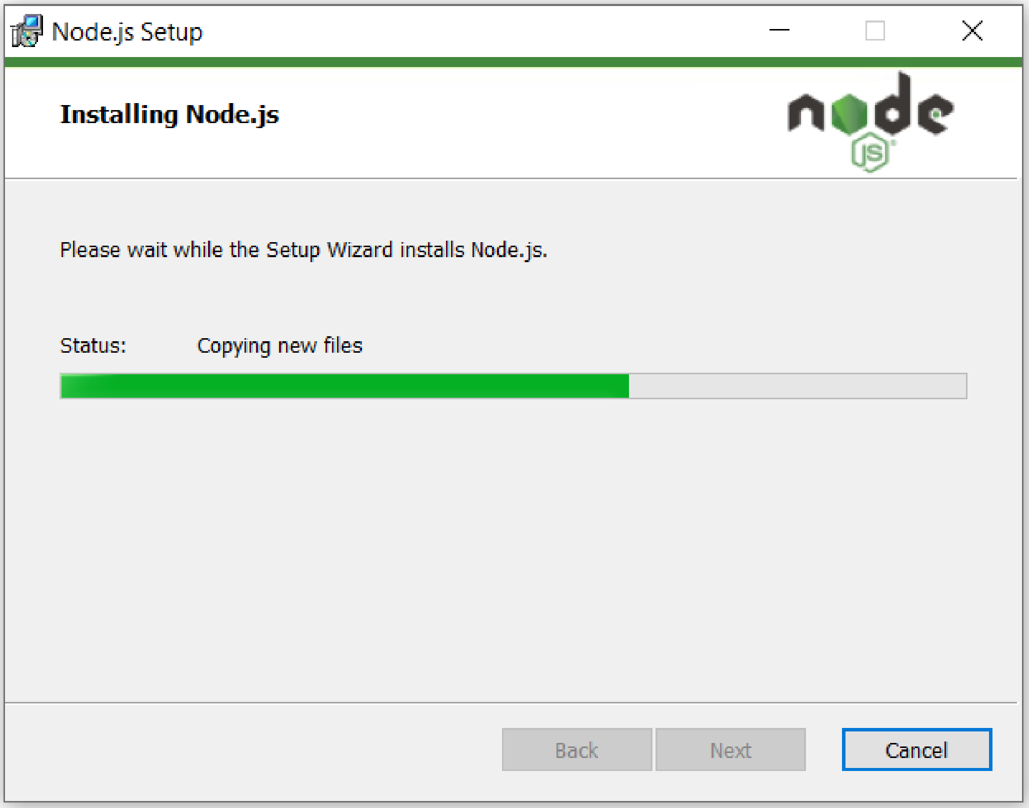
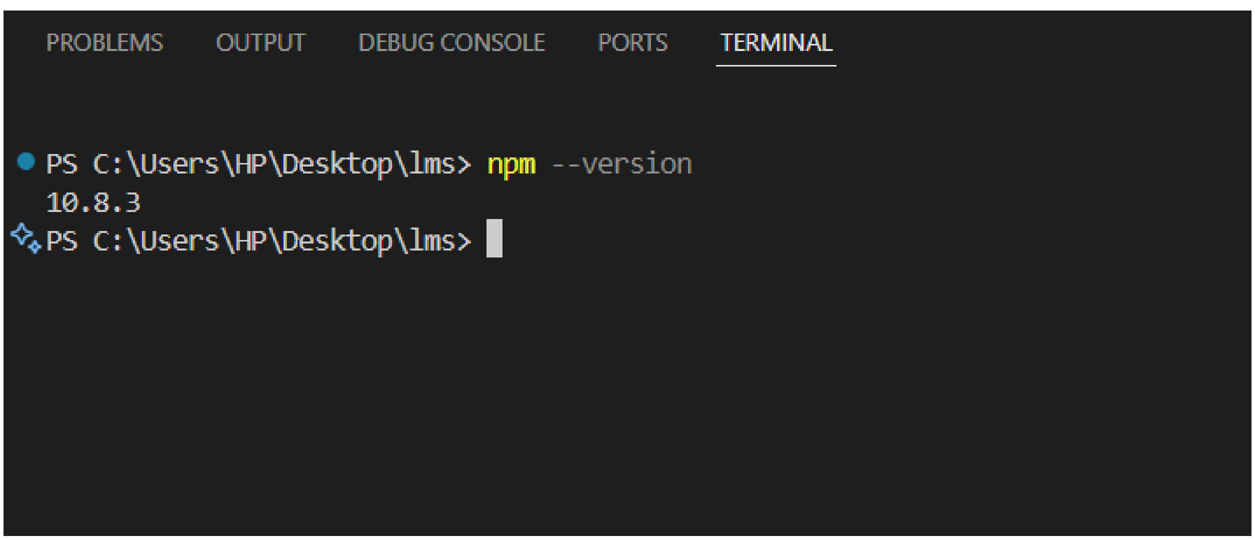
3. Angular Environment Setup





Step 1: Install Node.js

- Download from: https://nodejs.org
- Choose **LTS version**

Verify installation:

node -v

npm -v

Step 2: Install Code Editor

Recommended: **Visual Studio Code**

Useful VS Code extensions:

- Angular Language Service
- TypeScript Hero
- Prettier
- ESLint

Step 3: Browser

- Google Chrome (best for Angular debugging)
- Install **Angular DevTools** Chrome extension

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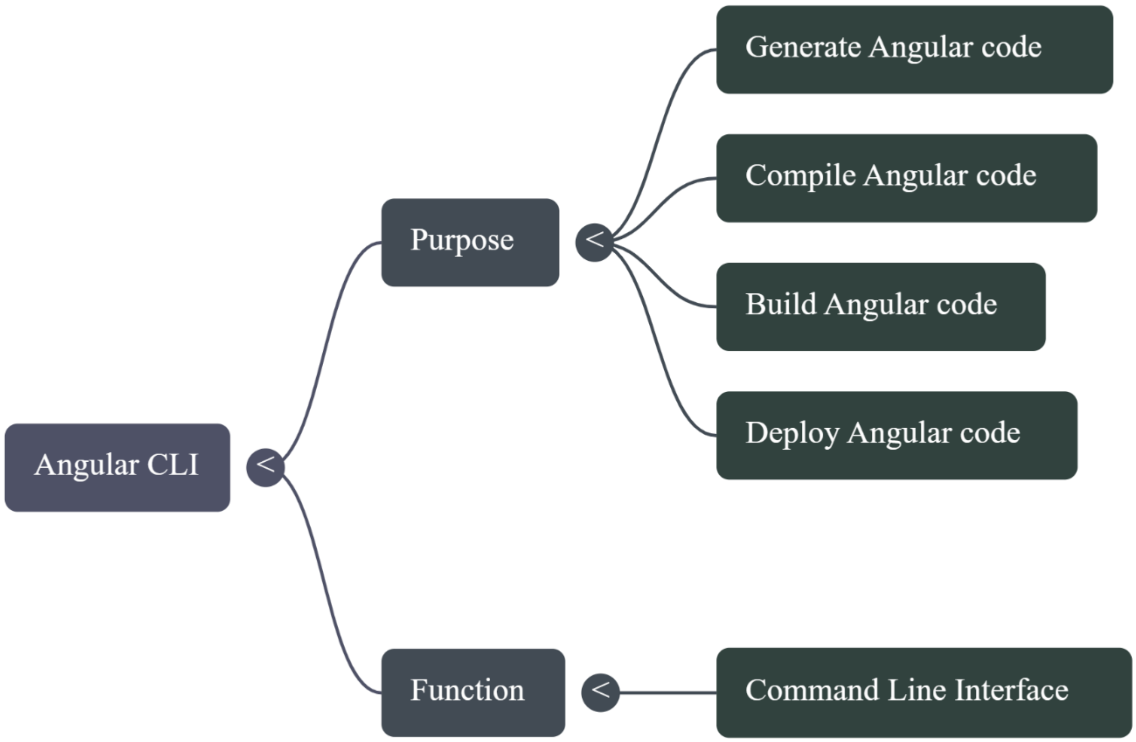
4. Angular CLI (Command Line Interface)



What is Angular CLI?

Angular CLI is a **powerful command-line tool** that:

- Creates projects
- Generates components/services
- Builds applications
- Runs development server
- Executes tests

Install Angular CLI

npm install -g @angular/cli

Verify:

ng version

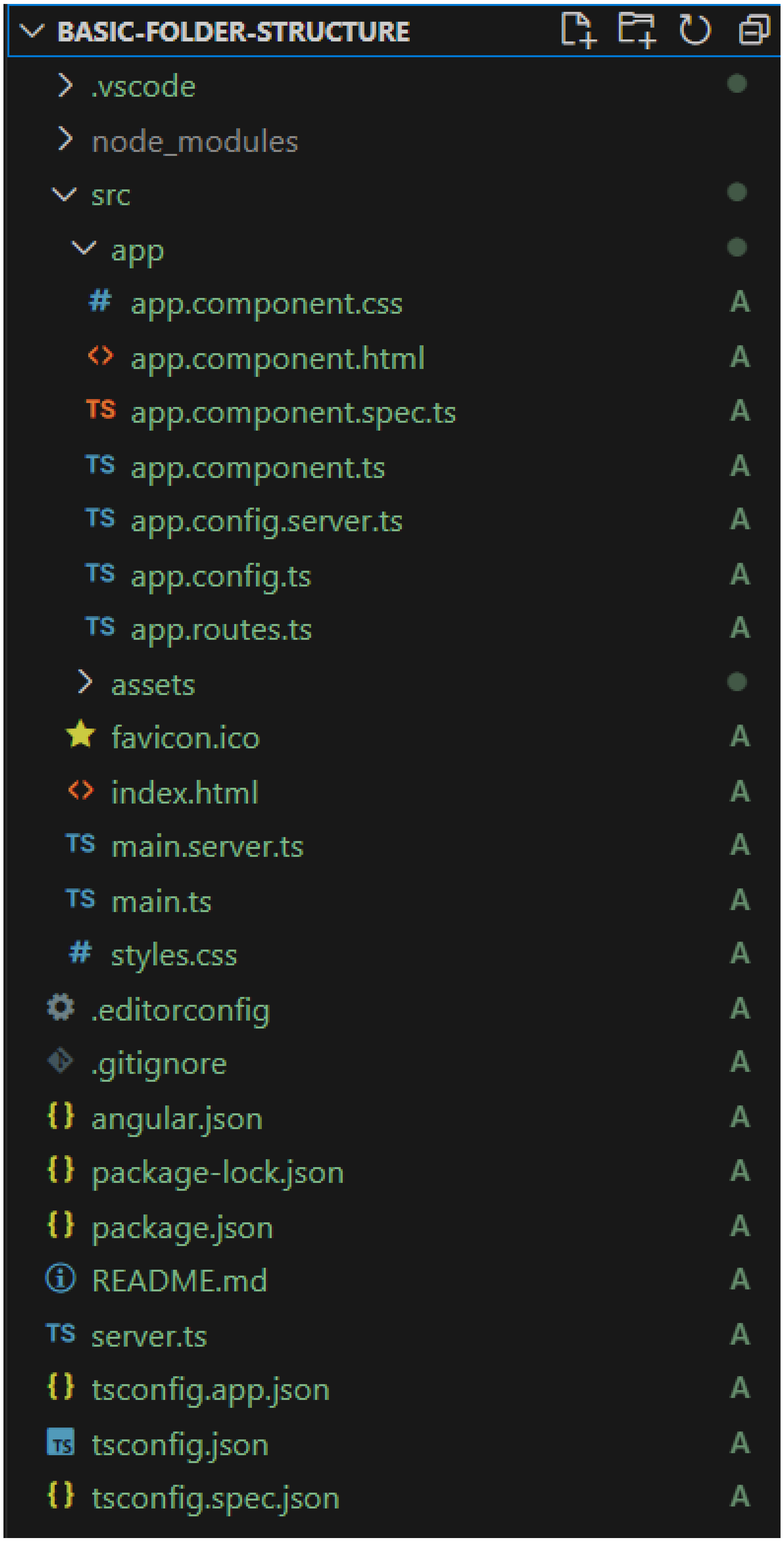
Common Angular CLI Commands

Command	**Purpose**
ng new	Create project
ng serve	Run app
ng build	Build for production
ng generate component	Create component
ng test	Unit testing
ng lint	Code quality

Example:

ng generate component product-list

5. Create Your First Angular Project



Step 1: Create Project

ng new my-first-angular-app

CLI will ask:

- Add routing? → Yes
- Stylesheet format? → CSS (or SCSS)

Step 2: Project Structure

```
my-first-angular-app
|— src
|   ├— app
|   |   ├— app.component.ts
|   |   ├— app.component.html
|   |   ├— app.component.css
|   |   └— app.module.ts
|   ├— assets
|   ├— index.html
|   └— main.ts
|— angular.json
|— package.json
```

Key Files Explained

- **main.ts** → Application entry point
- **app.module.ts** → Root module
- **app.component.ts** → Root component logic
- **app.component.html** → UI template
- **angular.json** → Angular build configuration

Step 3: Run Application

cd my-first-angular-app

ng serve

Open browser:

http://localhost:4200

You will see the **Angular welcome page**.

How Angular Application Works (Request Flow)

1. Browser loads index.html
2. main.ts bootstraps the app
3. AppModule loads
4. AppComponent renders UI
5. Angular handles routing & data binding

Summary

- Angular is a modern, enterprise-ready SPA framework
- Angular CLI simplifies development significantly
- Component-based architecture ensures scalability
- Environment setup is straightforward with Node + CLI
- First Angular app can be created in minutes

Next Recommended Topics

If you want, I can continue with:

1. Angular Architecture (Modules, Components, Services)
2. Data Binding (One-way & Two-way)
3. Directives & Pipes
4. Routing & Navigation
5. Angular CRUD with REST API (ASP.NET Core backend)

Just tell me how you want to proceed.